

Route decision: Shanghai Global Distribution Center → Rotterdam Distripark

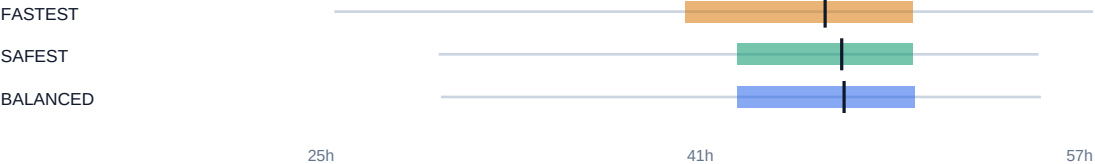
HUB-SHANGHAI → HUB-ROTTERDAM · mode any (STRICT) · cargo general · priority normal · scenarios: none · planned 2026-09-29 13:47 UTC · cargo value INR 960,699,450 at 25%/yr carrying cost

Recommendation: BALANCED, AIR via direct. Expected arrival in **42 h** (1.7 days); plan for 46 h (p85), worst case 49 h (p95). Cost: freight INR 12,842,895, landed INR 13,991,677 incl. INR 1,148,781 carrying cost. Peak leg risk 2%.

Options

Option	Mode	Via	Plan h	p50 h	p85 h	p95 h	Freight	Landed	Risk
FASTEST	AIR	direct	25	40	46	49	INR 13,331,606	INR 14,420,070	2%
SAFEST	AIR	direct	29	42	46	49	INR 13,390,771	INR 14,539,554	2%
BALANCED	AIR	direct	29	42	46	49	INR 12,842,895	INR 13,991,677	2%

ETA bands (bar p50–p95, tick p85, line plan → fully-correlated worst case)



Routes



Recommended route solid, alternatives dashed. Blue sea, amber air, green rail, orange road.

FASTEST: AIR via direct

AIR via no chokepoints; expected 39.7 h (p85 45.6 h, p95 49.3 h), INR 14,420,070 landed, 2 handoffs. 2.2 h faster than the next option. INR 428,376 more (landed) than BALANCED.

#	Mode	To	km	Hours	Delay p85	Cost	Risk	Signal
1	ROAD	Shanghai Pudong International	30	0.4	+3.6	INR 3,456	0%	
2	handoff	Shanghai Pudong International		6.0	+0.0	INR 14,410	2%	
3	AIR	Frankfurt Airport	8,858	11.1	+8.8	INR 12,764,824	0%	
4	AIR	Amsterdam Airport Cargo	366	0.5	+8.8	INR 527,124	0%	
5	handoff	Amsterdam Airport Cargo		6.0	+0.0	INR 14,410	2%	
6	ROAD	Rotterdam Distripark	64	0.9	+3.6	INR 7,381	0%	

Transit 12.9 h + transfers 12.0 h + announced scenario delay 0.0 h = plan 24.9 h. Model delay buffer p50/p85/p95: +14.7 / +24.9 / +32.0 h. Route band p50 39.7 h, p85 45.6 h, p95 49.3 h (independent legs); 56.9 h if all legs are late together. Costs: linehaul INR 13,302,785, transfers INR 28,821, disruption surcharge INR 0. Inventory carrying cost INR 1,088,464 expected (INR 1,351,669 at p95); landed INR 14,420,070.

Why the slowest leg (AIR-SHANGHAI → AIR-FRANKFURT, p85 8.8 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



SAFEST: AIR via direct

AIR via no chokepoints; expected 41.9 h (p85 46.3 h, p95 49.3 h), INR 14,539,554 landed, 2 handoffs. 2.2 h slower than FASTEST. INR 547,887 more (landed) than BALANCED.

#	Mode	To	km	Hours	Delay p85	Cost	Risk	Signal
1	ROAD	Shanghai Pudong International	30	0.4	+3.6	INR 3,456	0%	
2	handoff	Shanghai Pudong International		6.0	+0.0	INR 14,410	2%	
3	AIR	London Heathrow Airport	9,241	11.6	+8.8	INR 13,316,973	0%	
4	handoff	London Heathrow Airport		6.0	+0.0	INR 14,410	2%	
5	ROAD	Calais Channel Tunnel Hub	169	2.5	+3.6	INR 19,498	0%	
6	ROAD	Rotterdam Distripark	191	2.8	+3.6	INR 22,025	0%	

Transit 17.3 h + transfers 12.0 h + announced scenario delay 0.0 h = plan 29.3 h. Model delay buffer p50/p85/p95: +12.6 / +19.8 / +25.4 h. Route band p50 41.9 h, p85 46.3 h, p95 49.3 h (independent legs); 54.6 h if all legs are late together. Costs: linehaul INR 13,361,952, transfers INR 28,821, disruption surcharge INR 0. Inventory carrying cost INR 1,148,781 expected (INR 1,351,669 at p95); landed INR 14,539,554.

Why the slowest leg (AIR-SHANGHAI → AIR-HEATHROW, p85 8.8 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



BALANCED: AIR via direct

AIR via no chokepoints; expected 41.9 h (p85 46.4 h, p95 49.4 h), INR 13,991,677 landed, 2 handoffs. 2.2 h slower than FASTEST. INR 428,376 cheaper (landed) than the lowest-cost alternative.

#	Mode	To	km	Hours	Delay p85	Cost	Risk	Signal
1	ROAD	Shanghai Pudong International	30	0.4	+3.6	INR 3,456	0%	
2	handoff	Shanghai Pudong International		6.0	+0.0	INR 14,410	2%	
3	AIR	Frankfurt Airport	8,858	11.1	+8.8	INR 12,764,824	0%	
4	handoff	Frankfurt Airport		6.0	+0.0	INR 14,410	2%	
5	ROAD	Duisburg Intermodal Terminal	200	2.9	+3.6	INR 23,101	0%	
6	ROAD	Rotterdam Distripark	197	2.9	+3.6	INR 22,694	0%	

Transit 17.4 h + transfers 12.0 h + announced scenario delay 0.0 h = plan 29.4 h. Model delay buffer p50/p85/p95: +12.6 / +19.8 / +25.4 h. Route band p50 41.9 h, p85 46.4 h, p95 49.4 h (independent legs); 54.7 h if all legs are late together. Costs: linehaul INR 12,814,074, transfers INR 28,821, disruption surcharge INR 0. Inventory carrying cost INR 1,148,781 expected (INR 1,354,410 at p95); landed INR 13,991,677.

Why the slowest leg (AIR-SHANGHAI → AIR-FRANKFURT, p85 8.8 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



Method: legs are priced with gradient-boosted quantile models (p50/p85/p95) trained on synthetic delays fitted to published UNCTAD, World Bank and STB statistics; FASTEST plans on p50, BALANCED on p85 (plus cargo carrying cost when a value is given), SAFEST on p95 plus 240 risk-hours per unit of threat. Route bands assume independent legs. Sea distances are great-circle between chokepoint waypoints. Currency conversion: live (open.er-api.com) as of Tue, 29 Sep 2026 00:02:31 +0000.